

The Effects of Adenine Nucleotides on NADH Binding to Mitochondrial Malate Dehydrogenase¹

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The effects of adenine nucleotides on initial velocity and NADH binding have been studied with the malate dehydrogenase reaction. ATP, ADP, and AMP were inhibitors competitive with NADH and uncompetitive with oxaloacetate but caused only 50-60% inhibition at saturating concentrations. Direct fluorescence titrations indicated that saturating concentrations of the adenine nucleotides displaced 50-60% of the bound NADH from enzyme-NADH complex. Adenine and adenosine had no inhibitory effect but ADP-ribose caused complete inhibition and NADH dissociation. The possible mechanistic basis for these results and their physiological implications are discussed.

The relationship between kinetic phenomena and possible regulatory mechanisms of mitochondrial malate dehydrogenase is of considerable interest due to its central role in metabolism. Kinetic properties of the enzyme have been studied (1) indicating a compulsory binding order in both reaction directions, with coenzyme bound prior to substrate. Many unusual phenomena, such as substrate activation by malate (2) and inhibition by oxaloacetate (3), have been reported. Studies with ketomalonate as substrate demonstrated anomalous results, which were explained by the proposal of an alternating active site hypothesis (4, 5).

Adenine nucleotides have been reported to activate the oxidation of malate and inhibit the reduction of oxaloacetate (6). The activation was due to a decreased K_s for NAD^+ while the inhibition was competitive with NADH. These effects were abolished by treatment of the enzyme with HgCl_2 . Similar effects have been reported with the enzyme from *E. coli* (7) which were attributed to an allosteric site for NADH in addition to the catalytic site.

The present studies were undertaken in an attempt to characterize further the effects of adenine nucleotides on mitochondrial MDH. Fluorescence techniques, in addition to kinetic studies, provided evidence that the inhibition of oxaloacetate reduction by adenine nucleotides was not due to simple competition for the catalytic NADH binding site.

MATERIALS AND METHODS

Materials. Oxaloacetate, coenzymes, and adenine nucleotides were purchased from Sigma Chemical Corporation. The NADH was Sigma Grade III and was used without further purification. Its concentration was determined by absorbance at 340 nm, and 10 μM solutions were used to calibrate the fluorimeter for kinetic studies. D-Malate and hydroxymalonate were from Aldrich Chemical Company. Hydroxymalonate was twice recrystallized from ethanol prior to use. The pig heart mitochondrial malate dehydrogenase was either prepared in our laboratory or purchased from Servae Laboratories, Maidenhead, England. Commercial enzyme (Batch 82HX) exhibited maximum fluorescence emission at 330 nm and was further purified on DEAE-cellulose by the procedure of Chan and Schellenberg (8), yielding an enzyme with an emission spectrum identical to tyrosine. A modified procedure of Wolfe and Neilands (2) was used to prepare enzyme in our laboratory. Phos-

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phate buffer of pH 7.4, 0.1 M, was used instead of pH 7.0, 0.05 M. Maleate buffer containing 0.2 M NaCl and the zinc chloride treatment were omitted. DEAE-cellulose chromatography was substituted for the batchwise treatment with carboxymethylcellulose and electrophoretic isolation. This preparation procedure also gave enzyme with an emission spectrum identical to tyrosine. The enzyme was assayed by determining the initial rate of NADH oxidation at 338 m μ using 200 μ M oxaloacetate and NADH in 0.1 M phosphate buffer at pH 7.4.

The enzyme concentration was determined as NADH binding sites (9). The natural fluorescence of NADH is enhanced when it is bound to enzyme, and further enhanced in the presence of hydroxymalonate due to ternary complex formation. Approximately 5 μ N enzyme in the presence of 5 mM hydroxymalonate was titrated with NADH. The titrations were performed at pH 6.4 in 0.05 M acetate-Tris buffer, since it has been reported that the dissociation constant of NADH from the ternary complex is 0.05 μ M under these conditions (9).

Initial velocities of oxaloacetate reduction were measured by following the decrease in NADH fluorescence in an Eppendorf photometer with a fluorescence attachment. All the rate assays and fluorescence studies were performed in pH 8.0 acetate-Tris buffer, 0.05 M with respect to acetate, at 25°C. In the kinetic experiments the reaction was initiated by the addition of suitably diluted MDH, and the results were expressed as e/v in seconds. The ordinate intercepts of Figs. 1-3 do not represent maximum velocity since the concentration of oxaloacetate was not saturating. All kinetic studies were run in triplicate, and the data points in Figs. 1-4 and 6-8 represent the means of three determinations.

Fluorescence studies. The fluorescence spectra of enzyme, NADH and enzyme-NADH binary complex were studied using a Farrand recording spectrofluorometer. The decrease in fluorescence of enzyme-NADH complex due to adenine nucleotides was monitored at fixed optimum excitation and emission wavelengths. A Corning CS7-54 primary filter and 10-nm slits were used.

The decrease of fluorescence was expressed as a fractional change of the difference of fluorescence between free and bound NADH. The expression

$$(\Delta F I_i - \Delta F I) / \Delta F I_i$$

was used to calculate fractional change in fluorescence. The $\Delta F I_i$ represented the difference in fluorescence between free and enzyme-bound NADH while $\Delta F I$ represented the decrease in fluorescence of the binary complex due to the addition of adenine nucleotides. Appropriate

enzyme and buffer blanks were subtracted from observed values.

RESULTS

Figures 1, 2, and 3 demonstrate that ATP, ADP and AMP are all inhibitors kinetically competitive with NADH. Similar studies with adenine and adenosine did not show an inhibitory effect. The inhibition due to ATP with respect to oxaloacetate is demonstrated in Fig. 4. Although these reciprocal plots were parallel, the inhibition expected

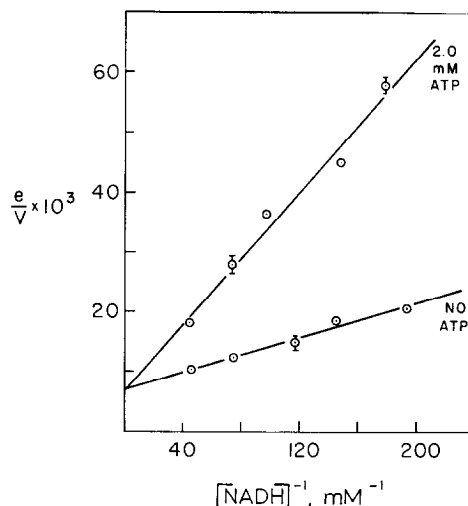


Fig. 1. Double-reciprocal plot of initial velocity vs NADH concentration in the absence and presence of 2 mM ATP. The oxaloacetate concentration was 0.1 mM.

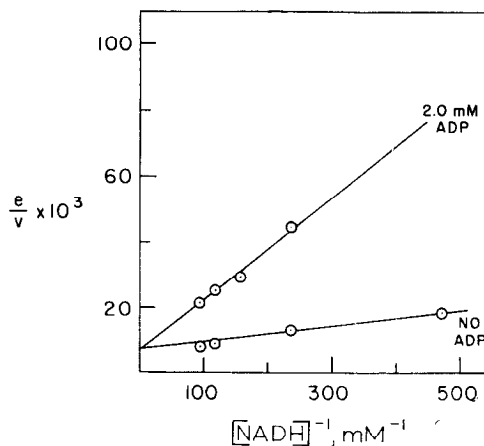


Fig. 2. Double-reciprocal plot of initial velocity vs NADH concentration in the absence and presence of 2 mM ADP. The oxaloacetate concentration was 0.1 mM.

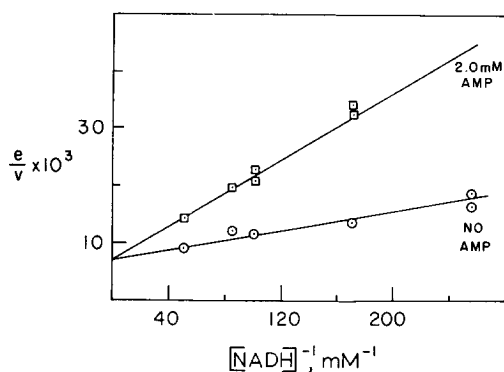


FIG. 3. Double-reciprocal plot of initial velocity vs NADH concentration in the absence and presence of 2 mM AMP. The oxaloacetate concentration was 0.1 mM.

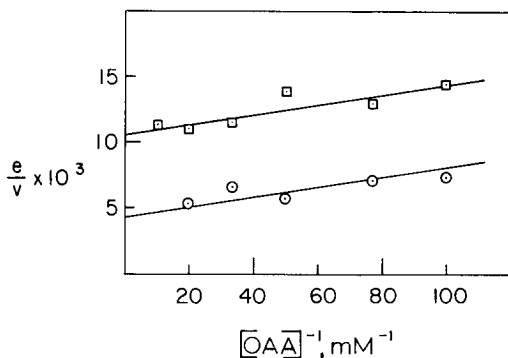


FIG. 4. Double-reciprocal plot of initial velocity vs oxaloacetate concentration in the absence $\circ$ and presence $\square$ of 1.5 mM ATP. The NADH concentration was 20 μ M.

with this enzyme, which follows an ordered bi-bi mechanism, should be noncompetitive (10). The uncompetitive inhibition obtained is probably only apparent, due to weak substrate binding and a limited range of substrate and inhibitor concentrations. Nevertheless, the changed intercept obtained is compatible with an inhibitor competitive for coenzyme. Similar results were obtained with AMP and ADP. Attempts to study the inhibition competitive with NADH using the method of Dixon (11), however, resulted in hyperbolic curves of reciprocal velocity versus adenine nucleotide concentration in the range of 0–3.0 mM ATP. Consequently, the direct effects of these ligands on the fluorescence of binary enzyme–NADH complex were studied.

The binding of NADH to mitochondrial malate dehydrogenase resulted in a small blue shift and approximately 2-fold enhancement of the NADH fluorescence emission spectrum. Figure 5 shows the spectrum of free NADH, enzyme-bound NADH, and the results of successive additions of ATP. The fluorescence intensity was diminished by adenine nucleotide addition to a minimal level at 1.7 mM ATP. Subsequent additions resulted in a negligible decrease in fluorescence and it was therefore not possible to reduce the spectrum to the level of free NADH. To determine whether the NADH had been modified during the experiment, oxaloacetate was added resulting in complete oxidation of bound and free NADH.

The incomplete dissociation of binary enzyme–NADH complex due to ATP was investigated for AMP and ADP with identical results. In addition, the ratios of inhibited to uninhibited velocities were determined at various concentrations of each of the adenine nucleotides since the diminished coenzyme binding should result in a decrease in reaction velocity. The result of a typical study is presented in Fig. 6, in which both fractional velocity and fractional displacement of NADH from binary complex are plotted against ATP concentration. It can be seen that ATP did not cause complete inhibition or complete dissociation of binary complex, even at a concentra-

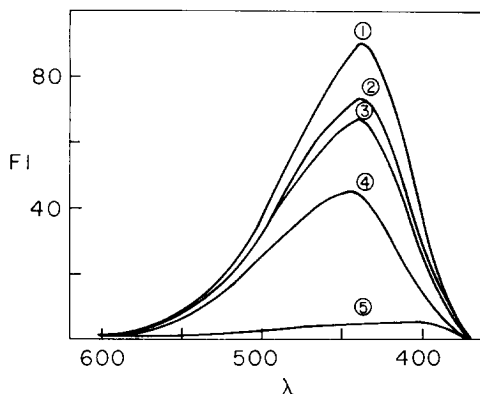


FIG. 5. Fluorescence emission Spectrum 1, 9.2 μ M *m*-MDH–NADH complex excited at 345 m μ ; Spectra 2 and 3 are after addition of 0.1 mM and 1.7 mM ATP, respectively; Spectrum 4, 9.2 μ M free NADH; Spectrum 5, the fluorescence emission after addition of 0.1 mM oxaloacetate.

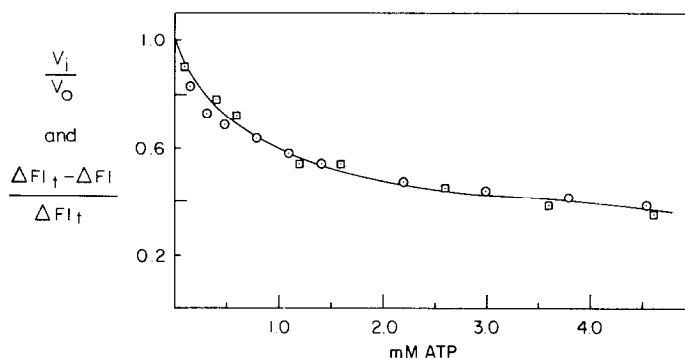


FIG. 6. Fractional fluorescence change $\circ$ and ratio of inhibited to uninhibited velocity $\square$ at various ATP concentrations. Fluorescence titration was performed with $11.4 \mu\text{M}$ *m*-MDH under the conditions of Figure 5. The rate assays were carried out with $11.4 \mu\text{M}$ NADH and $100 \mu\text{M}$ oxaloacetate.

tion of 4.6 mM. Similar results were obtained with ADP and AMP. In all cases, the curves reached a plateau at a ratio of 0.4–0.5, possibly indicating displacement from half the binding sites.

In order to investigate structural requirements for these unusual effects, ADP-ribose was studied. Figure 7 shows the competitive inhibition obtained, from which a K_I of $114 \mu\text{M}$ was calculated. In Fig. 8 the fractional fluorescence changes and V_i/V_0 ratios approach the x axis, indicating that the inhibitor is able to totally displace NADH from the binary complex. Thus, ADP-ribose can effectively compete for all the NADH binding sites in contrast with the adenine nucleotides. Adenosine and adenine were also screened but showed no inhibitory effects up to a concentration of 5 mM.

In order to more accurately determine the fractional activity and fractional fluorescence at extrapolated infinite adenine nucleotide concentration, the data of Figs. 6 and 8 were plotted as suggested by Webb (12). In this plot the kinetic data were plotted as $[1 - (V_i/V_0)]^{-1}$ and the fluorescence titration data as $\{1 - [(\Delta F_l - \Delta F_i)/\Delta F_l]\}^{-1}$, both versus reciprocal adenine nucleotide concentration. The concentration range used was 0–2.5 mM, since at very high adenine nucleotide concentrations the inhibition appears to become more complex. The data of Fig. 8 resulted in an ordinate intercept of 1.0 at extrapolated infinite ADP-ribose, as expected for com-

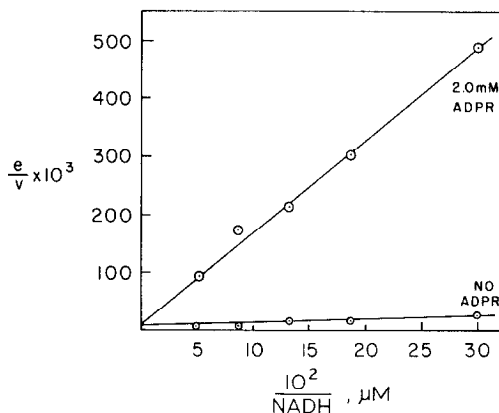


FIG. 7. Double-reciprocal plot of initial velocity vs NADH concentrations in the absence and presence of 2 mM ADPR. The experimental conditions were similar to those described in Fig. 1.

pletely competitive inhibition. The data of Fig. 6 resulted in an ordinate intercept of 1.6, which would be expected for partial competitive inhibition. This value corresponds to a limiting value for V_i/V_0 , or fractional fluorescence change, of 0.375 at extrapolated infinite ATP.

DISCUSSION

The double-reciprocal plots in the presence and absence of adenine nucleotides indicate that their inhibitory effect is competitive with NADH. At extrapolated infinite concentrations of coenzyme either the adenine nucleotides are no longer bound or they are still bound but do not affect the activity

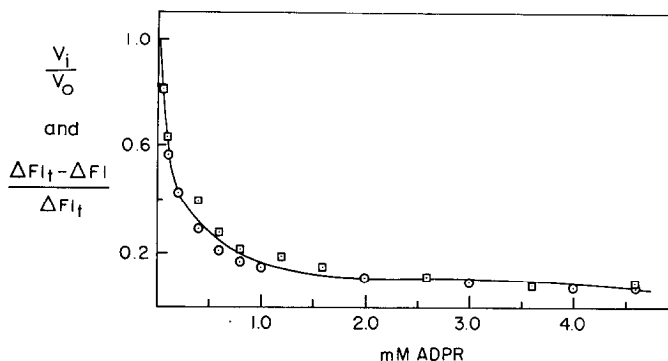


FIG. 8. Fractional fluorescence change $\circ$ and ratio of inhibited to uninhibited velocity $\square$ at various ADPR concentrations. Fluorescence titration was performed with $9.3 \mu\text{M}$ *m*-MDH and the rate assays with $12.0 \mu\text{M}$ NADH. Other experimental conditions were identical to those described in Fig. 6.

of the enzyme. At increasing concentrations of adenine nucleotides, however, complete inhibition is not seen and the ratio of inhibited to uninhibited velocity seems to reach a minimum at approximately 0.4.

The difference in fluorescence emission due to binding NADH to the enzyme is comparable to that reported by Theorell and Langan (13). Additions of adenine nucleotides were unable to reduce the fluorescence emission signal of binary enzyme-NADH complex to that of free NADH, and the curve of fractional fluorescence change at various concentrations of inhibitor correlated with the V_i/V_0 curve. The complete oxidation of free and bound NADH by oxaloacetate addition indicates that the failure of saturating adenine nucleotides to reduce the emission to the level of free NADH is not due to a modification of the coenzyme. Thus, both the kinetic and fluorescence data indicate incomplete dissociation of the binary enzyme-coenzyme complex by saturating adenine nucleotides.

The type of inhibition caused by adenine nucleotides can be classified as partial competitive (14) but the mechanism by which this inhibition is obtained is still not clear. The hyperbolic nature of the Dixon plots suggest a complex mechanism in which adenine nucleotides bind to more than one enzyme form. Binding of the adenine nucleotides at a site other than the coenzyme binding site could cause either a conformational change or subunit dissociation result-

ing in decreased affinity for the coenzyme. Another possibility would be ligand exclusion (15) in which the adenine nucleotides could displace only the adenine ring of coenzyme, resulting in decreased affinity but enabling excess coenzyme to displace the adenine nucleotide. The fact that adenine nucleotides can displace only approximately half the bound NADH is compelling and may indicate asymmetry of the two subunits in the dimeric enzyme when it is complexed with NADH. This could be related to the alternating active site hypothesis previously proposed for the enzyme (5). A worthwhile model might be the recently reported evidence that only one subunit of the crystalline supernatant enzyme binds NAD^+ (16). Extensive further studies, however, would be required to delineate clearly the exact mechanism responsible for the unusual effects of adenine nucleotides.

With regard to the possible physiological role of the adenine nucleotides, the concentration required for substantial effects on activity are within the realm of localized intramitochondrial levels. The complete displacement of coenzyme by ADP-ribose indicates that the unique aspect of adenine nucleotide inhibition is abolished as the molecular structure more closely approximates that of NADH. Since similar effects were found for all three nucleotides tested it is difficult to assign a specific regulatory role for them based on the present results. Future studies, including additional screen-

ing of metabolites, may help to clarify both the regulatory mechanism and its physiological implications.

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